MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 01 (SOLUTIONS)
Chapter 3: Pair of Linear Equations in Two Variables

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions
MATHS MANTHAN | (3 × 1 = 3 Marks)

Q1. The pair of linear equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ has: [1]

(A) a unique solution (B) exactly two solutions (C) infinitely many solutions (D) no solution

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(D) no solution** [0.5 Mark]

Given: $a_1 = 1,\ b_1 = 2,\ c_1 = 5$ and $a_2 = -3,\ b_2 = -6,\ c_2 = 1$. We apply the consistency ratio test:

Formula / Theorem Used: If $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$, the lines are parallel and the system has **no solution**.

Justification: [0.5 Mark]

$\frac{a_1}{a_2} = \frac{1}{-3} = -\frac{1}{3}$, $\frac{b_1}{b_2} = \frac{2}{-6} = -\frac{1}{3}$, $\frac{c_1}{c_2} = \frac{5}{1} = 5$. Since $-\frac{1}{3} = -\frac{1}{3} \neq 5$, there is **no solution**.

Q2. If a pair of linear equations in two variables is consistent, then the lines represented by them are: [1]

(A) parallel (B) always coincident (C) intersecting or coincident (D) always intersecting

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) intersecting or coincident** [0.5 Mark]

We apply the geometric criterion of consistency:

Formula / Theorem Used: A linear system is **consistent** iff it possesses at least one common solution.

Justification: [0.5 Mark]

Intersecting lines meet at one unique point (consistent), while coincident lines overlap at infinitely many points (dependent consistent). Parallel lines never intersect (inconsistent). Thus lines are **intersecting or coincident**.

Q3. The pair of equations $y = 0$ and $y = -7$ has: [1]

(A) one solution (B) two solutions (C) infinitely many solutions (D) no solution

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(D) no solution** [0.5 Mark]

We know the standard geometric property of horizontal lines:

Formula / Theorem Used: Equations $y = k_1$ and $y = k_2$ $(k_1 \neq k_2)$ represent distinct horizontal lines parallel to the x-axis.

Justification: [0.5 Mark]

$y = 0$ is the x-axis and $y = -7$ is parallel to it 7 units below. Since they never intersect (also $0 \neq -7$), they have **no solution**.

SECTION B: Assertion-Reason Question MATHS MANTHAN | (1 × 1 = 1 Mark)

Q4. **Assertion (A):** The system of linear equations $9x + 3y + 12 = 0$ and $18x + 6y + 24 = 0$ has infinitely many solutions.

Reason (R): A pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ has infinitely many solutions if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification of (A) & (R): [0.5 Mark]

Comparing coefficients: $a_1 = 9, b_1 = 3, c_1 = 12$ and $a_2 = 18, b_2 = 6, c_2 = 24$.

$$\frac{a_1}{a_2} = \frac{9}{18} = \frac{1}{2}, \qquad \frac{b_1}{b_2} = \frac{3}{6} = \frac{1}{2}, \qquad \frac{c_1}{c_2} = \frac{12}{24} = \frac{1}{2}$$

Since $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{2}$, the lines are coincident and have infinitely many solutions. Hence both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION C: Very Short Answer Questions MATHS MANTHAN | (2 × 2 = 4 Marks)

Q5. On comparing the ratios $\frac{a_1}{a_2}$, $\frac{b_1}{b_2}$ and $\frac{c_1}{c_2}$, determine whether the pair of equations $5x - 8y + 1 = 0$ and $\frac{3}{2}x - \frac{12}{5}y + \frac{1}{3} = 0$ is consistent or inconsistent. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying Coefficients & Computing Ratios [1 Mark]

Comparing with $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$: $a_1 = 5$, $b_1 = -8$, $c_1 = 1$ and $a_2 = \frac{3}{2}$, $b_2 = -\frac{12}{5}$, $c_2 = \frac{1}{3}$.

Formula / Theorem Used: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies$ Parallel Lines $\implies$ No Solution (Inconsistent)

$$\frac{a_1}{a_2} = \frac{5}{3/2} = \frac{5 \times 2}{3} = \frac{10}{3}, \qquad \frac{b_1}{b_2} = \frac{-8}{-12/5} = \frac{8 \times 5}{12} = \frac{40}{12} = \frac{10}{3}, \qquad \frac{c_1}{c_2} = \frac{1}{1/3} = 3$$

Step 2: Comparing Ratios & Conclusion [1 Mark]

Since $\frac{10}{3} = \frac{10}{3} \neq 3$, we have $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. The pair represents parallel lines with no solution.

The pair of linear equations is inconsistent.

Q6. Solve the following pair of linear equations using the substitution method:

$$x + y = 14 \quad \text{and} \quad x - y = 4$$

[2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $x + y = 14$ — (i) and $x - y = 4$ — (ii)

Step 1: Expressing x in terms of y & Substituting [1 Mark]

From equation (ii), transposing $-y$ to RHS: $x = 4 + y$ — (iii)

Substituting $x = 4 + y$ into equation (i):

$$(4 + y) + y = 14 \implies 4 + 2y = 14 \implies 2y = 14 - 4 = 10 \implies y = \frac{10}{2} = 5$$

Step 2: Back-Substitution to Find x [1 Mark]

Putting $y = 5$ in (iii): $x = 4 + 5 = 9$.

$$\boxed{\mathbf{x = 9, \quad y = 5}}$$

SECTION D: Short & Long Answer Questions MATHS MANTHAN | ($1 \times 3 + 1 \times 5 = 8$ Marks)

Q7. Solve the following pair of linear equations graphically:

$$2x + y = 6 \quad \text{and} \quad 2x - y + 2 = 0$$

Find the coordinates of the vertices of the triangle formed by these lines and the x-axis. Also, calculate its area. [3]

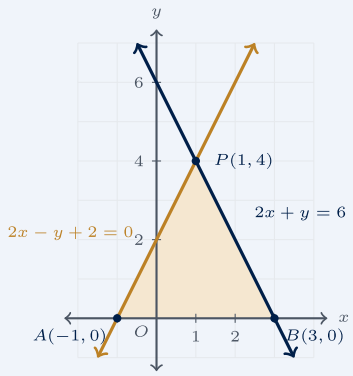
▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Solution Tables for Both Lines [1 Mark]

For $2x + y = 6 \implies y = 6 - 2x$ and $2x - y + 2 = 0 \implies y = 2x + 2$:

x	0	3	1
$y = 6 - 2x$	6	0	4

x	-1	0	1
$y = 2x + 2$	0	2	4



Step 2: Graphical Intersection & Vertices [1 Mark]

Plotting both lines, they intersect at $P(1, 4)$. Hence the unique solution is $x = 1, y = 4$.

Vertices of $\triangle PAB$ formed with the x-axis ($y = 0$): $P(1, 4)$, $A(-1, 0)$, and $B(3, 0)$.

Step 3: Area of $\triangle PAB$ [1 Mark]

Formula / Theorem Used: Area of $\triangle = \frac{1}{2} \times \text{Base} \times \text{Height}$

Base $AB = 3 - (-1) = 4$ units, Height $= y_P = 4$ units.

$$\text{Area}(\triangle PAB) = \frac{1}{2} \times 4 \times 4 = 8 \text{ sq. units}$$

Solution: $(\mathbf{1}, \mathbf{4})$; Vertices: $(\mathbf{-1}, \mathbf{0})$, $(\mathbf{3}, \mathbf{0})$, $(\mathbf{1}, \mathbf{4})$; Area = **8** sq. units

Q8. A train covered a certain distance at a uniform speed. If the train had been 10 km/h faster, it would have taken 2 hours less than the scheduled time. And, if the train were slower by 10 km/h, it would have taken 3 hours more than the scheduled time. Find the distance covered by the train. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let uniform speed be x km/h and scheduled time be y hours.

Formula / Theorem Used: Distance (D) = Speed $\times$ Time $= xy$

Step 1: Framing Linear Equations from Both Conditions [2 Marks]

Case I: $(x+10)(y-2) = xy \implies xy - 2x + 10y - 20 = xy \implies -2x + 10y = 20 \implies -x + 5y = 10$ — (i)

Case II: $(x-10)(y+3) = xy \implies xy + 3x - 10y - 30 = xy \implies 3x - 10y = 30$ — (ii)

Step 2: Solving for x and y by Elimination [2 Marks]

Multiplying equation (i) by 2: $-2x + 10y = 20$ — (iii)

Adding (ii) and (iii) to eliminate y:

$$(3x - 10y) + (-2x + 10y) = 30 + 20 \implies x = 50 \text{ km/h}$$

Substituting $x = 50$ in (i): $-50 + 5y = 10 \implies 5y = 60 \implies y = 12$ hours.

Step 3: Calculating Total Distance Covered [1 Mark]

$$\text{Distance} = x \times y = 50 \times 12 = \boxed{\mathbf{600}\text{ km}}$$

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Q9. CASE STUDY: Architectural Layout of an Apartment (Ravish's Floor Plan)

Ravish is planning to buy a 2-BHK apartment whose layout is designed on a coordinate grid. The total area of two identical bedrooms and the kitchen together is 95 m^2. The length of each bedroom is x m and width is 5 m. The length of the kitchen is y m and width is 5 m. The front outer wall of the apartment has a total length of 15 m consisting of Bedroom 1, a 2 m wide bathroom, and the kitchen. [4]

(i) Formulate the pair of linear equations in x and y representing the outer length and the combined area of the two bedrooms and kitchen. [1]

(ii) Find the length of each bedroom (x) and the length of the kitchen (y). [1]

(iii) Calculate the total area of the living room. [1]

(iv) Find the cost of laying floor tiles in the kitchen at the rate of Rs. 50 per m^2. [1]

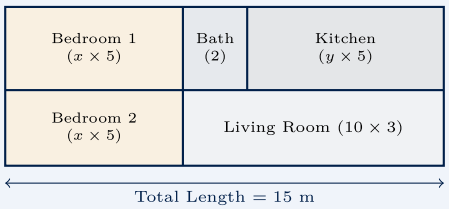
— OR —

Find the total perimeter of the combined two bedrooms. [1]

► CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Extracted Scenario Data: Two bedrooms each $x \times 5$ m^2, bath width 2 m, kitchen $y \times 5$ m^2, living room 10×3 m^2 (per floor plan), total front length = 15 m.

Formula / Theorem
Used: Area $= l \times b$ and Perimeter $= 2(l + b)$



Sub-part (i): Formulating Linear Equations [1 Mark]

Front outer wall: $x + 2 + y = 15 \implies x + y = 13$ — (i)

Combined area: $2(5x) + 5y = 95 \implies 10x + 5y = 95 \implies \boxed{\mathbf{2x + y = 19}}$ — (ii) (and $\mathbf{x + y = 13}$)

Sub-part (ii): Finding x and y [1 Mark]

Subtracting (i) from (ii): $(2x + y) - (x + y) = 19 - 13 \implies x = 6$ m.

From (i): $6 + y = 13 \implies y = 7$ m. Hence $\boxed{\text{Bedroom length } \mathbf{x = 6} \text{ m, Kitchen length } \mathbf{y = 7} \text{ m}}$.

Sub-part (iii): Total Area of the Living Room [1 Mark]

From the labeled floor plan dimensions (10 m × 3 m): Area $= 10 \times 3 = \boxed{\mathbf{30\ m^2}}$.

Sub-part (iv): Cost of Laying Kitchen Floor Tiles [1 Mark]

Kitchen Area $= y \times 5 = 7 \times 5 = 35$ m^2. Cost at Rs. $50/m^2 = 35 \times 50 = \boxed{\text{Rs. } \mathbf{1{,}750}}$.

Sub-part (iv) [OR Option]: Total Perimeter of Combined Two Bedrooms [1 Mark]

Combined vertically along the shared wall of length $x = 6$ m, the two bedrooms form a 6 m × (5 + 5) m = 6 m × 10 m rectangle:

$$\text{Perimeter} = 2(6 + 10) = 2 \times 16 = \boxed{\mathbf{32\ m}} \quad (\text{sum of individual rooms } = 2 \times 2(6 + 5) = 44 \text{ m})$$

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